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PUBLICATION LIST

Papers and Preprints

33. Dong Wang and Thomas Wolfs, Invariant pseudo-unitary matrix ensembles and local statistics and conductance fluctuations of the DMPK ensemble, `https://arxiv.org/abs/2610.27532`.

32. Zhigang Bao, Dong Wang and Yue Zhu, Eigenvector distribution of random matrices under critical finite-rank deformations, `https://arxiv.org/abs/2605.30779`.

31. Dong Wang and Shuai-Xia Xu, Critical Hermitian matrix model with external source and Boussinesq hierarchy, `https://arxiv.org/abs/2512.20343`.

30. Dong Wang and Shuai-Xia Xu, Hard to soft edge transition for the Muttalib-Borodin ensembles with integer parameter θ, `https://arxiv.org/abs/2501.16746`.

29. Dong Wang, Hard edge universality of Muttalib-Borodin ensembles with real parameter θ, `https://arxiv.org/abs/2312.14840`.

28. Dong Wang and Dong Yao, Biorthogonal polynomials related to quantum transport theory of disordered wires, *Comm. Math. Phys.*, 407 (2026), no. 24, 79pp.

27. Tom Claeys and Dong Wang, Universality for random matrices with equi-spaced external source: a case study of a biorthogonal ensemble, *J. Stat. Phys*, 188 (2022) no. 2, paper no. 11, 28 pp.

26. Dong Wang and Lun Zhang, A vector Riemann-Hilbert approach to the Muttalib-Borodin ensembles, *J. Funct. Anal.*, 282 (2022), no. 7, 109380, 84 pp.

25. Swapnil Yadav, Kazi Alam, Khandker Muttalib and Dong Wang, Non-monotonic confining potential and eigenvalue density transition for generalized random matrix model, *Phys. Rev. E*, 103 (2021), no. 4, 042137, 11 pp.

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21. Dang-Zheng Liu, Dong Wang and Yanhui Wang, Lyapunov exponent, universality and phase transition for products of random matrices, *Comm. Math. Phys.*, 399 (2023), 1811–1855.

20. Karl Liechty and Dong Wang, Asymptotics of free fermions in a quadratic well at finite temperature and the Moshe-Neuberger-Shapiro random matrix model, *Ann. Inst. Henri Poincaré Probab. Stat.*, 56 (2020), no. 2, 1072–1098.

19. Eunghyun Lee and Dong Wang, Distributions of a particle's position and their asymptotics in the q-deformed totally asymmetric zero range process with site dependent jumping rates, *Stochastic Process. Appl.*, 129 (2019), no. 5, 1795–1828.

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17. Tom Claeys, Arno B. J. Kuijlaars, Karl Liechty and Dong Wang, Propagation of singular behavior for Gaussian perturbations of random matrices, *Comm. Math. Phys.*, 362 (2018), no. 1, 1–54.

16. Karl Liechty and Dong Wang, Two Lax systems for the Painlevé II equation, and two related kernels in random matrix theory, *SIAM J. Math. Anal.*, 48 (2016), no. 5, 3618–3666.

15. Dong Wang and David Waugh, The transition probability of the q-TAZRP (q-Bosons) with inhomogeneous jump rates, *SIGMA* 12 (2016), 036, 16 pp, Contribution to the special issue on Asymptotics and Universality in Random Matrices, Random Growth Processes, Integrable Systems and Statistics in honor of Percy Deift and Craig Tracy.

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11. Ivan Corwin, Zhipeng Liu and Dong Wang, Fluctuations of TASEP and LPP with general initial data, *Ann. Appl. Probab.*, 26 (2016), no. 4, 2030–2082.

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7. Jinho Baik and Dong Wang, On a relationship between high rank cases and rank one cases of Hermitian random matrix models with external source, *Random Matrix Theory, Interacting Particle Systems and Integrable Systems*, Edited by Percey Deift and Peter Forrester, MSRI Publications 65 (2014), Cambridge University Press, Cambridge, 25–38.

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1. Dong Wang, A PDE for the multi-time joint probability of the Airy process, *Phys. D*, 238 (2009), no. 8, 819–833.

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